Sovereign Post-Quantum Cryptographic Enclaves and Transformation Optics for Cross-Border EHDS & GDPR Secondary Health Vaults

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ABSTRACT

ABSTRACT: Cross-border secondary use of clinical health records within the European Health Data Space (EHDS) requires mathematical non-malleability, strict GDPR Article 9 data subject anonymity, and compliance with BSI TR-02102-1 and ANSSI post-quantum directives. In this work, we integrate number-theoretic transformation optics with cyclotomic quotient ring cryptography R_q = Z_q[X]/(X^256 + 1) using NIST FIPS 203 (ML-KEM-1024) and FIPS 204 (ML-DSA-87). By coupling Ramanujan trigonometrical sums c_q(n) and Dirichlet character impedance matching, we achieve full physical and cryptographic data erasure with zero mutual information leakage.

Boundary reflection at the cloaking interface is completely suppressed by imposing Dirichlet character boundary conditions and Ramanujan sum orthogonality:

$$c_q(n) = \sum_{d \mid \gcd(n,q)} d\mu(q/d), \quad \chi(a) = \exp\left(2\pi i \frac{a \cdot \mathrm{ind}(a)}{q}\right)$$

European cybersecurity regulations demand unconditional quantum resistance against Shor's and Grover's algorithms:

$$\mathrm{Adv}^{\mathrm{IND-CCA2}}_{\mathrm{ML-KEM-1024}}(\mathcal{A}) \le \mathrm{Adv}^{\mathrm{M-LWE}}_{R_q,\beta}(\mathcal{B}) + 2^{-256}, \quad \varepsilon_{\mathrm{total}} = \sum_{i=1}^{m} \frac{\alpha \Delta_2^2}{2\sigma_i^2} + \frac{\ln(1/\delta)}{\alpha - 1}$$

1. Number-Theoretic Invisibility via Ramanujan Trigonometrical Sums

2. European Regulatory and Sovereign Framework Alignment

- EU GDPR (Arts 9, 25, 32): Data protection by design with provable k-anonymity (k=50) and differential privacy.
- EHDS Regulation: Cross-border secondary use enclave isolation with cryptographic zero-telemetry.
- BSI TR-02102-1 (Germany): FIPS 203 compliant ML-KEM-1024 parameters for long-term health archive security.
- ANSSI Directives (France): Hybrid post-quantum migration strategy using deterministic recurrent-prime ladders.

3. Empirical EHDS Enclave Performance & EU Regulatory Audit







